LEARNING MODULE

11

Eigen-decomposition and PCA

The directions in which a covariance matrix only stretches: how many separate movements 28 Taiwan sectors have, how stable they are, and why the inverse matrix amplifies error.

VOLUME II · Linear Algebra for Portfolios

BUILDS ON LM7 · LM8 · LM9 · LM10

LEADS TO LM12 · LM16 · LM20 · LM22 · LM23 · LM24 · LM27 · LM28 · LM30 · LM43 · LM45

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** calculate the eigenvalues and eigenvectors of a 2×2 symmetric matrix by hand, and of an $N \times N$ one with `eigh`
☐	**2** explain the spectral theorem and its conditions, and write a covariance matrix as $\hat{\Sigma} = Q\Lambda Q^\top = \sum_k \lambda_k \mathbf{q}_k \mathbf{q}_k^\top$
☐	**3** derive the first principal component as the unit vector of largest variance, with the eigenvalue as its Lagrange multiplier
☐	**4** interpret explained shares, loadings and scores, and compare covariance with correlation PCA
☐	**5** explain the sign convention for eigenvectors and why near-equal eigenvalues make their eigenvectors unstable
☐	**6** compare principal components across calendar years and halves of a sample, and label the comparison a stability check
☐	**7** calculate the condition number $\kappa = \lambda_1/\lambda_N$ and explain why solving a system with $\hat{\Sigma}$ amplifies estimation error

LEARNING MODULE OVERVIEW

- An eigenvector of M is a direction that M only stretches, $M\mathbf{x} = \lambda_k \mathbf{x}$. The daily covariance matrix of Semiconductor and FinancialInsurance stretches the direction 34.2° from the Semiconductor axis by 3.0492 and the perpendicular one by 0.7559, in units of 10^{-4}.
- A real symmetric matrix has real eigenvalues and orthonormal eigenvectors, so $\hat{\Sigma} = \sum_k \lambda_k \mathbf{q}_k \mathbf{q}_k^\top$. Each eigenvalue is the variance in the direction of its eigenvector, and every unit direction has a variance between λ_N and λ_1.
- The first principal component maximizes variance over unit vectors, and its Lagrange multiplier is the eigenvalue. Over 2008–2026 one direction carries 52.6% of the standardized variance of 28 Taiwan sector indices, but 12 directions are needed to pass 80%.
- Covariance PCA follows the most volatile sectors and changes with units; correlation PCA treats every sector alike. Restating Semiconductor in percent turns its covariance PC1 into Semiconductor alone, with a share of 99.996%.
- An aggregate index beside its members creates a near-zero eigenvalue: the correlation matrix of Electronic and its eight sub-industries has $\lambda_9 = 0.0073$ and condition number 836.
- A perturbation rotates eigenvectors by about its size divided by the eigenvalue gap. The first component barely moves between years (cosine at least 0.98); beyond it, the components of the two halves of the sample change places (second cosine 0.297).
- The inverse weights each direction by $1/\lambda_k$. Between the two halves U28's covariance matrix moved by 24.5% in the spectral norm and the solution of $\hat{\Sigma}\mathbf{x} = \mathbf{1}$ by 0.729 of the second half's solution length, against a worst case of 30.93.

1 INTRODUCTION

A covariance matrix of 28 sectors has 406 distinct entries (LM10). Its structure lies in the perpendicular directions that it only stretches, its eigenvectors, and in the stretch factors, its eigenvalues. They rank the data's uncorrelated movements by variance and explain why the inverse matrix magnifies small changes.

MOTIVATING CASE **Twenty-eight sectors, how many movements?**

Take U28, the 28 leaf indices among the 30 Taiwan Stock Exchange sector indices that FinMind serves daily: all but the aggregates Electronic and ChemicalBiotechnologyMedicalCare. From 2 December 2008 to 8 October 2026 their daily returns are all positively correlated, from 0.217 (BiotechnologyMedicalCare and OilGasElectricity) to 0.755 (ComputerPeripheralEquipment and ElectronicPartsComponents).

One fixed mix of the 28 standardized returns, with entries from 0.130 to 0.224, carries 52.6% of their total variance. Yet 12 such directions are needed to pass 80%, and the first one's share was 66.6% in 2020 but 34.5% in 2017. A second direction is positive on the eight electronics sub-industries and on BiotechnologyMedicalCare, Chemical and ElectricMachinery, and negative on the other 17.

On 7 April 2025 all 28 fell, by 5.90% (CommunicationsInternet) to 9.95% (Semiconductor). Yet over the sample all 28 moved the same way on only 300 of 4,382 days (6.8%).

Solved on each half of the sample, the linear system $\hat{\Sigma}\mathbf{x} = \mathbf{1}$ moves Semiconductor's entry from 430.84 to −587.60 and FinancialInsurance's from −2,547.30 to 2,216.20 (Exhibit 1). How many separate movements are there, do they hold over time, and why does the solution move so much more than the matrix? Sections 5, 7 and 8 answer.

Exhibit 1: Twenty-eight Taiwan sector indices, 2 December 2008 to 8 October 2026

Quantity	Value
Daily returns	4,382
Pairwise correlations (378): smallest, mean, largest	0.217, 0.502, 0.755
Share of the first direction (correlation matrix)	52.6%
Directions needed to pass 80%	12
First direction's share by calendar year: lowest, highest	34.5% (2017), 66.6% (2020)
Days on which all 28 rose; all 28 fell	144; 156
7 April 2025: smallest and largest fall	5.90%, 9.95%
$\hat{\Sigma}\mathbf{x} = \mathbf{1}$, Semiconductor: first half, second half	430.84, −587.60
$\hat{\Sigma}\mathbf{x} = \mathbf{1}$, FinancialInsurance: first half, second half	−2,547.30, 2,216.20
Relative change $\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\| / \|\mathbf{x}^{(2)}\|$	0.729

Source: FinMind `TaiwanStockPrice`, daily closes of the 28 sector price indices of U28, retrieved 11 October 2026, with 524 closes on 38 dates repaired from the 13:30:00 values of `TaiwanStockEvery5SecondsIndex`. The halves are the first and the last 2,191 returns, split after 6 October 2017; $\hat{\Sigma}$ has divisor T and daily units. Computations: `data/lm11_numbers.py`.

Sections 2–4 build the tools: eigenvectors, the spectral theorem and the variance in any direction. Sections 5–7 turn them into principal component analysis and test its stability. Section 8 explains the moving solution, and Section 9 does the work in Python.

> **NOTE** **Notation for this module**
>
> As in LM10: $\mathbf{R}_t$ the N daily simple returns on date t ($R_{j,t}$ that of asset j), $\bar{\mathbf{R}}$ their mean, $\hat{\Sigma}$ the sample covariance matrix (divisor T), $\mathbf{s}$ the standard deviations, C the correlation matrix, $A = 245.48$ sessions a year on the U28 sample. A generic matrix is M, entries m_{ij}, a system $M\mathbf{x} = \mathbf{b}$; eigenvalues $\lambda_1 \geq \cdots \geq \lambda_N$, unit eigenvectors $\mathbf{q}_k$, $Q = (\mathbf{q}_1 \ \cdots \ \mathbf{q}_N)$, $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_N)$, condition number $\kappa = \lambda_1/\lambda_N$. As in LM7: Σ the annualized covariance matrix of 0050 and 00679B, with entries σ_1^2, σ_{12}, σ_2^2. Local: ℓ the unknown of the characteristic equation (often λ, but here λ is a Lagrange multiplier and the λ_k are the roots; not LM9's elimination multipliers ℓ_{ik}); $\mathbf{u}$ a unit vector; $h(\mathbf{u})$ the Rayleigh quotient (often R or ρ, here a return and a correlation); $\xi_{k,t}$ a score (not $\mathbf{z}_t$, LM12's whitened returns); $\bar{\rho}$ the mean off-diagonal entry of C; λ without subscript a Lagrange multiplier (ζ a second one, since the book reserves μ, ν and φ), $\mathcal{L}$ the Lagrangian; $^{(1)}$, $^{(2)}$ the two halves of the sample; ΔM, $\Delta\mathbf{x}$ changes of a matrix and a vector; $\|M\|_2$ the spectral norm; ε a small off-diagonal entry; θ an angle; $\mathbf{y}$, a a generic vector and number. Counts: N assets, T returns; i, j assets, t dates, k and k' components.

> **NOTE** **Research hygiene**
>
> The universe (U28) and the sample were fixed before any result was computed. Full-sample eigenvalues, shares, loadings and condition numbers are **descriptive**: they summarize the past, and nobody inside the sample could have known them. Comparisons across calendar years and across the two halves, split by time, are a **stability check**, not an out-of-sample test. A low cosine between two periods cannot tell noise from a real change.

2 DIRECTIONS A MATRIX ONLY STRETCHES

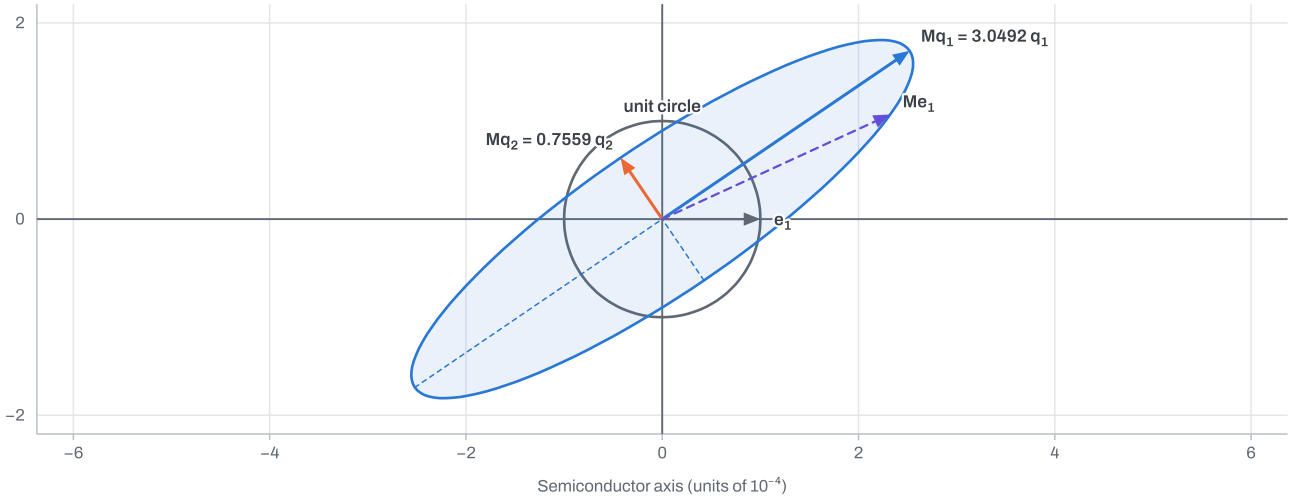
- [ ] calculate the eigenvalues and eigenvectors of a 2×2 symmetric matrix by hand, and of an $N \times N$ one with `eigh`

Write $M := 10^4\hat{\Sigma}$ for the daily returns of Semiconductor and FinancialInsurance on the U28 sample:

$$M = \begin{pmatrix} 2.3234 & 1.0666 \\ 1.0666 & 1.4817 \end{pmatrix}.$$

Multiply every unit vector $\mathbf{u}$ by M. The unit circle becomes an ellipse (Exhibit 2), and most vectors also turn: $\mathbf{e}_1$ goes to $(2.3234,\ 1.0666)^\top$, which points 24.7° away. Two perpendicular directions do not turn. Along them M only stretches, by 3.0492 along one and by 0.7559 along the other.

Exhibit 2: The unit circle and its image under a covariance matrix



Note: $M = 10^4\hat{\Sigma}$ for Semiconductor (horizontal axis) and FinancialInsurance (vertical axis). The two eigenvectors keep their directions under M, while $\mathbf{e}_1$ turns. Computations: `data/lm11_charts.py`.

DEFINITION **Eigenvalue and eigenvector**

Let M be an $N \times N$ matrix. A number λ_k and a vector $\mathbf{x} \neq \mathbf{0}$ with $M\mathbf{x} = \lambda_k \mathbf{x}$ are an **eigenvalue** and an **eigenvector** of M, together an **eigenpair**. The eigenvector keeps its direction under M and is stretched by the factor λ_k, reversed if $\lambda_k < 0$.

Rewrite the definition as $(M - \lambda_k I)\mathbf{x} = \mathbf{0}$. A non-zero solution exists exactly when $M - \lambda_k I$ is singular (LM9, (9.6)), so

$$\begin{aligned} & M\mathbf{x} = \lambda_k \mathbf{x} \ \text{ for some } \mathbf{x} \neq \mathbf{0} \\ \iff & \det(M - \lambda_k I) = 0. \end{aligned} \tag{11.1}$$

The second line is the **characteristic equation**. For a 2×2 matrix and an unknown ℓ it is a quadratic, with the **trace** $\operatorname{tr} M = m_{11} + m_{22}$:

$$\begin{aligned} 0 &= \det(M - \ell I) \\ &= (m_{11} - \ell)(m_{22} - \ell) - m_{12}m_{21} \\ &= \ell^2 - (\operatorname{tr} M)\,\ell + \det M. \end{aligned}$$

The quadratic formula gives both roots:

$$\lambda_{1,2} = \tfrac{1}{2}\operatorname{tr} M \pm \sqrt{\tfrac{1}{4}(\operatorname{tr} M)^2 - \det M}. \tag{11.2}$$

Factored, the quadratic is $(\ell - \lambda_1)(\ell - \lambda_2)$. So **the eigenvalues add up to the trace and multiply to the determinant**: $\lambda_1 + \lambda_2 = \operatorname{tr} M$ and $\lambda_1\lambda_2 = \det M$. Section 3 extends both facts to symmetric matrices of any size.

A rotation by an angle other than 0° or 180° turns every vector of the plane, so it has no real eigenvector (Strang 2016, §6.1, Example 5). Symmetry excludes such matrices.

EXAMPLE 1

The eigenpairs of a two-sector covariance matrix

Find the eigenvalues and unit eigenvectors of $M = 10^4\hat{\Sigma}$ for Semiconductor and FinancialInsurance, with $m_{11} = 2.3234$, $m_{22} = 1.4817$ and $m_{12} = 1.0666$.

SOLUTION

The trace is 3.8051, so $\frac{1}{2}\operatorname{tr} M = 1.90255$. For a symmetric matrix the term under the root of (11.2) equals $\frac{1}{4}(m_{11} - m_{22})^2 + m_{12}^2$ (Section 3), here $0.17711 + 1.13764 = 1.31475$, with root 1.14663. So $\lambda_1 = 3.0492$ and $\lambda_2 = 0.7559$. They add up to the trace, and their product 2.3049 is the determinant $2.3234 \times 1.4817 - 1.0666^2$.

For λ_1, the first row of $(M - \lambda_1 I)\mathbf{x} = \mathbf{0}$ reads $-0.7258\,x_1 + 1.0666\,x_2 = 0$. So $x_2/x_1 = 0.6805$, an angle of 34.2° from the Semiconductor axis, and the unit eigenvector is $\mathbf{q}_1 = (0.827,\ 0.563)^\top$. The second is perpendicular to it (Section 3), $\mathbf{q}_2 = (-0.563,\ 0.827)^\top$.

PITFALL **An eigenvalue is not a sector's variance**

Here $\lambda_1 = 3.0492$ exceeds both variances, 2.3234 and 1.4817, and $\lambda_2 = 0.7559$ is below both. Each eigenvalue is the variance of a mix along its eigenvector (Section 4). The mix 0.827 Semiconductor plus 0.563 FinancialInsurance has 1.31 times the variance of Semiconductor alone.

KNOWLEDGE CHECK 1

A symmetric 2×2 matrix has trace 5 and determinant 4. Find its eigenvalues without taking a square root.

ANSWER

The eigenvalues add up to 5 and multiply to 4, so they are 4 and 1, the roots of $\ell^2 - 5\ell + 4 = 0$.

3 SYMMETRIC MATRICES: THE SPECTRAL THEOREM

- ☐ explain the spectral theorem and its conditions, and write a covariance matrix as $\hat{\Sigma} = Q\Lambda Q^\top = \sum_k \lambda_k \mathbf{q}_k \mathbf{q}_k^\top$

Covariance and correlation matrices are symmetric, $m_{ij} = m_{ji}$, and symmetry buys three properties.

THEOREM **Spectral theorem**

Let M be a real symmetric $N \times N$ matrix. Then (i) all its eigenvalues are real; (ii) eigenvectors of different eigenvalues are orthogonal; (iii) one can choose mutually orthogonal unit eigenvectors $\mathbf{q}_1, \dots, \mathbf{q}_N$, so that $Q^\top Q = I$ and

$$M = Q \Lambda Q^\top = \sum_{k=1}^{N} \lambda_k \, \mathbf{q}_k \mathbf{q}_k^\top . \tag{11.3}$$

Proof for $N = 2$. (i) The term under the root of (11.2) is

$$\begin{aligned} &\tfrac{1}{4}(m_{11} + m_{22})^2 - (m_{11} m_{22} - m_{12}^2) \\ &\quad = \tfrac{1}{4}(m_{11} - m_{22})^2 + m_{12}^2 \;\geq 0, \end{aligned}$$

so both roots are real. They are equal only when $m_{11} = m_{22}$ and $m_{12} = 0$, that is, when M is a multiple of I.

(ii) Let $M\mathbf{x}_1 = \lambda_1 \mathbf{x}_1$ and $M\mathbf{x}_2 = \lambda_2 \mathbf{x}_2$ with $\lambda_1 \neq \lambda_2$, and read $\mathbf{x}_1^\top M \mathbf{x}_2$ two ways. Multiplying $M\mathbf{x}_2$ first gives $\lambda_2 \, \mathbf{x}_1^\top \mathbf{x}_2$. By symmetry $\mathbf{x}_1^\top M = (M\mathbf{x}_1)^\top$, which gives $\lambda_1 \, \mathbf{x}_1^\top \mathbf{x}_2$. So $(\lambda_1 - \lambda_2)\, \mathbf{x}_1^\top \mathbf{x}_2 = 0$ and $\mathbf{x}_1^\top \mathbf{x}_2 = 0$, for any N.

(iii) Scale the eigenvectors to unit length; if $\lambda_1 = \lambda_2$, M is a multiple of I and any orthonormal pair will do. The columns of MQ are $\lambda_1 \mathbf{q}_1$ and $\lambda_2 \mathbf{q}_2$, so $MQ = Q\Lambda$. Orthonormal columns give $Q^\top Q = I$, hence $Q^{-1} = Q^\top$ and $M = Q \Lambda Q^\top$. Written as columns times rows, as in LM10's (10.2), this product is the sum in (11.3). ■

Section 5 proves (i) and (iii) for every N. Three consequences follow from (11.3).

Trace and determinant. The diagonal of $\mathbf{q}_k \mathbf{q}_k^\top$ sums to $\mathbf{q}_k^\top \mathbf{q}_k = 1$, so $\operatorname{tr} M = \sum_k \lambda_k$. Since $Q^{-1} = Q^\top$, also $QQ^\top = I$, and the product rule (9.3) gives $\det M = \prod_k \lambda_k$.

Definiteness. In the coordinates $\mathbf{y} = Q^\top \mathbf{x}$, (11.3) gives $\mathbf{x}^\top M \mathbf{x} = \sum_k \lambda_k y_k^2$. So M is positive semidefinite exactly when every $\lambda_k \geq 0$ (take $\mathbf{x} = \mathbf{q}_k$ for the converse), and positive definite exactly when every $\lambda_k > 0$. This is LM10's promised eigenvalue test (Strang 2016, §6.5), and a sample covariance matrix, positive semidefinite by (10.5), never has a negative eigenvalue.

Signs and ties. If $\mathbf{q}_k$ is a unit eigenvector, so is $-\mathbf{q}_k$. The book's sign rule makes the entry largest in absolute value positive, the first of them on a tie. A 2×2 correlation matrix with $\rho > 0$ thus has $\mathbf{q}_2 = (1, -1)^\top / \sqrt{2}$; for $\rho < 0$ the same vector is $\mathbf{q}_1$. When two eigenvalues are equal, every unit vector in the plane of their eigenvectors is an eigenvector. Orthonormal eigenvectors can then be chosen, but only the plane, the **eigenspace**, is determined.

EXAMPLE 2

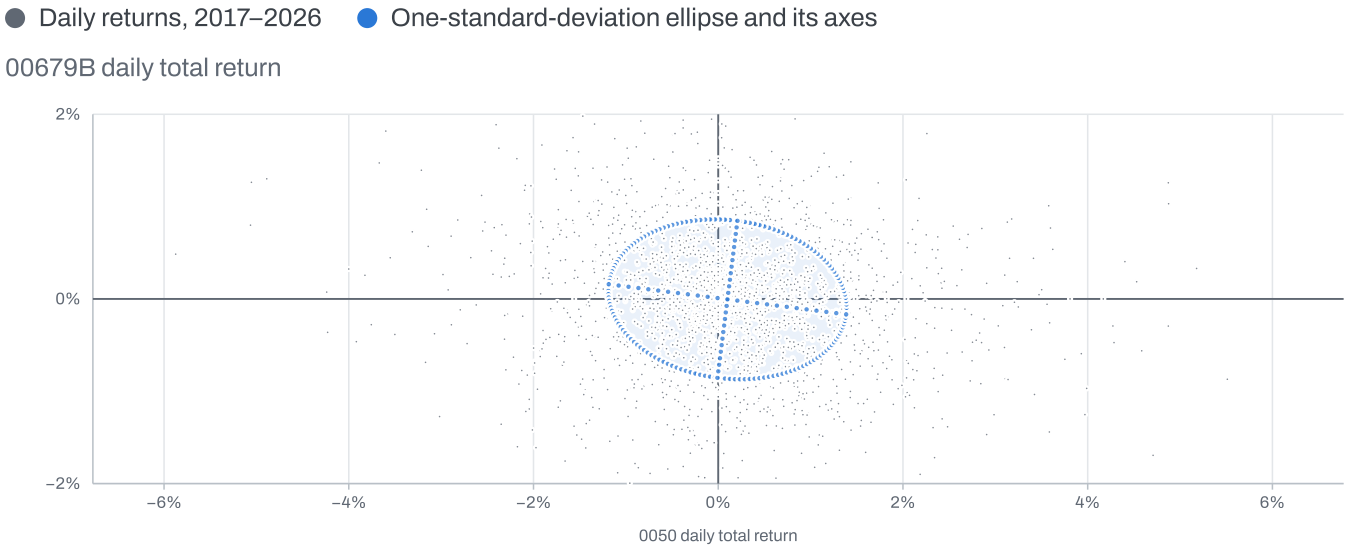
LM7's covariance matrix in its own axes

LM7 used the annualized covariance matrix $\Sigma = A\hat{\Sigma}$ of 0050 and 00679B over 2017–2026, with $A = 243.0543$ on its sample. Its entries are $\sigma_1^2 = 0.040425$, $\sigma_2^2 = 0.018214$ and $\sigma_{12} = -0.002866$, and LM7 quoted its eigenvalues without proof. Find the eigenpairs and the volatilities along the two axes.

SOLUTION

The trace is 0.058639, and (11.2) gives $\lambda_1 = 0.040789$ and $\lambda_2 = 0.017850$. The first row of $(\Sigma - \lambda_1 I)\mathbf{x} = \mathbf{0}$ gives $x_2/x_1 = (0.040789 - 0.040425)/(-0.002866) = -0.1270$, so $\mathbf{q}_1 = (0.992,\ -0.126)^\top$, tilted -7.24° from the 0050 axis. The axis volatilities, $\sqrt{\lambda_1} = 20.20\%$ and $\sqrt{\lambda_2} = 13.36\%$, are close to the funds' 20.11% and 13.50%: a correlation of −0.106 tilts the axes only slightly (Exhibit 3).

Exhibit 3: Daily total returns of 0050 and 00679B with the one-standard-deviation ellipse



Note: LM7's 2,363 daily total returns, 18 January 2017 to 8 October 2026; days with larger moves, such as 7 April 2025, lie outside the window. The ellipse is $(\mathbf{R} - \bar{\mathbf{R}})^\top \hat{\Sigma}^{-1} (\mathbf{R} - \bar{\mathbf{R}}) = 1$ for the daily $\hat{\Sigma}$; it describes the sample and is not a probability contour (LM16). Source: FinMind `TaiwanStockPrice` and `TaiwanStockDividendResult`, retrieved 10 October 2026; `data/lm11_charts.py`.

EXAMPLE 3

Three rank-one pieces

For S3 (Semiconductor, FinancialInsurance, Food) on the U28 sample, in units of 10^{-4},

$$10^4 \hat{\Sigma} = \begin{pmatrix} 2.3234 & 1.0666 & 0.6610 \\ 1.0666 & 1.4817 & 0.7795 \\ 0.6610 & 0.7795 & 1.1550 \end{pmatrix}.$$

The matrix's eigenvalues, from `eigh`, are $\lambda_1 = 3.4728$, $\lambda_2 = 0.9944$ and $\lambda_3 = 0.4929$, with signed eigenvectors in the columns of

$$Q = \begin{pmatrix} 0.7366 & 0.6526 & -0.1774 \\ 0.5492 & -0.4240 & 0.7202 \\ 0.3948 & -0.6279 & -0.6707 \end{pmatrix}.$$

Write the matrix as the sum (11.3), and find the first piece's share of the trace.

SOLUTION

Piece k, $\lambda_k \mathbf{q}_k \mathbf{q}_k^\top$, has entries $\lambda_k q_{ik} q_{jk}$, where q_{ik} is entry i of $\mathbf{q}_k$. To three decimals, computed from the printed eigenpairs,

$$k=1: \begin{pmatrix} 1.884 & 1.405 & 1.010 \\ 1.405 & 1.047 & 0.753 \\ 1.010 & 0.753 & 0.541 \end{pmatrix},$$
$$k=2: \begin{pmatrix} 0.424 & -0.275 & -0.407 \\ -0.275 & 0.179 & 0.265 \\ -0.407 & 0.265 & 0.392 \end{pmatrix},$$
$$k=3: \begin{pmatrix} 0.016 & -0.063 & 0.059 \\ -0.063 & 0.256 & -0.238 \\ 0.059 & -0.238 & 0.222 \end{pmatrix}.$$

Added entry by entry, they return $10^4 \hat{\Sigma}$ to within 0.001, the rounding of the pieces. The first piece has every covariance positive and carries $3.4728/4.9601 = 70.0\%$ of the trace $2.3234 + 1.4817 + 1.1550 = 4.9601$.

PITFALL **Comparing eigenvectors entry by entry**

An eigenvector's sign is a convention, and when two eigenvalues are close the vector itself can turn. Compare two periods' eigenvectors by $|\cos\theta|$, the absolute value of their dot product (Section 7), never entry by entry.

導讀 **對稱矩陣：沿互相垂直的方向拉伸**

矩陣把單位圓變成橢圓。大多數向量（vector）會被轉向，只有特徵向量（eigenvector）的方向不變，只被拉長或壓短，倍數就是特徵值（eigenvalue）。共變異數矩陣（covariance matrix）是對稱（symmetric）的，所以譜定理（spectral theorem）保證特徵值都是實數、特徵向量可以取成互相垂直，矩陣可以寫成幾個秩一（rank-one）矩陣的加總，這就是譜分解（spectral decomposition）。特徵值的和等於各變數變異數（variance）的總和；三個產業的例子中，第一個方向就占了 70.0%。特徵向量的正負號沒有意義，本書規定絕對值最大的元素為正；兩個特徵值相等時，只有它們張成的平面是確定的。

KNOWLEDGE CHECK 2

Is $-\mathbf{q}_k$ an eigenvector of M? What fixes the sign of $\mathbf{q}_k$ in this book?

ANSWER

Yes: $M(-\mathbf{q}_k) = -\lambda_k \mathbf{q}_k = \lambda_k(-\mathbf{q}_k)$, and $-\mathbf{q}_k$ also has unit length. The sign rule makes the entry largest in absolute value positive, the first one on a tie.

4 VARIANCE IN A DIRECTION: THE RAYLEIGH QUOTIENT AND THE CONDITION NUMBER

- ☐ derive the first principal component as the unit vector of largest variance, with the eigenvalue as its Lagrange multiplier
- ☐ calculate the condition number $\kappa = \lambda_1/\lambda_N$ and explain why solving a system with $\hat{\Sigma}$ amplifies estimation error

Read $\mathbf{u}^\top \mathbf{R}_t$, for a unit vector $\mathbf{u}$, as a mix of the N returns, rebalanced daily. Its sample variance is $\mathbf{u}^\top \hat{\Sigma} \mathbf{u}$ (LM10, (10.4)), the variance of the data in the direction $\mathbf{u}$. For any symmetric M and any $\mathbf{u} \neq \mathbf{0}$, the ratio

$$h(\mathbf{u}) := \frac{\mathbf{u}^\top M \mathbf{u}}{\mathbf{u}^\top \mathbf{u}}$$

is the **Rayleigh quotient**. Rescaling $\mathbf{u}$ does not change it, so unit vectors suffice.

PROPOSITION Bounds on the variance in a direction

Let M be symmetric with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_N$. For every unit vector $\mathbf{u}$,

$$\lambda_N \leq \mathbf{u}^\top M \mathbf{u} \leq \lambda_1, \tag{11.4}$$

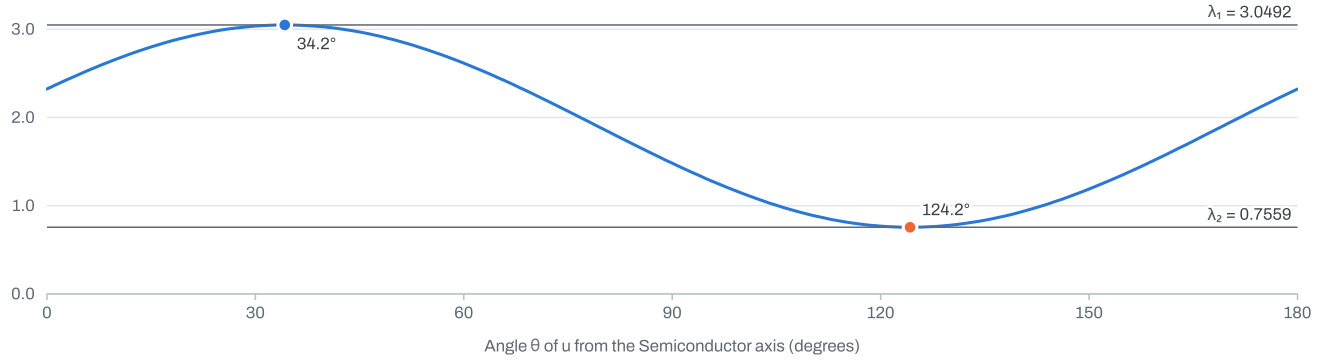
with equality on the right at $\mathbf{u} = \mathbf{q}_1$ and on the left at $\mathbf{u} = \mathbf{q}_N$.

Proof. Put $\mathbf{y} = Q^\top \mathbf{u}$. Since $QQ^\top = I$, $\|\mathbf{y}\|^2 = \mathbf{u}^\top Q Q^\top \mathbf{u} = 1$. By (11.3), $\mathbf{u}^\top M \mathbf{u} = \mathbf{y}^\top \Lambda \mathbf{y} = \sum_k \lambda_k y_k^2$: an average of the eigenvalues with weights y_k^2 that sum to 1. It lies between the smallest and the largest eigenvalue, and $\mathbf{u} = \mathbf{q}_1$ puts all the weight on λ_1. ■

LM7 used these bounds without proof: its smallest and largest curvatures of Σ are λ_2 and λ_1 of Example 2. Exhibit 4 traces $h(\mathbf{u})$ for Semiconductor and FinancialInsurance.

Exhibit 4: The variance of a two-sector mix in every direction

h(u), variance in direction u (units of 10^{-4})



Note: $h(\mathbf{u})$ for $\mathbf{u} = (\cos\theta, \sin\theta)^\top$ and the matrix $M = 10^4\hat{\Sigma}$ of Example 1. The maximum, 3.0492 at 34.2°, and the minimum, 0.7559 at 124.2°, are the eigenvalues.

The condition number. For a positive definite M, the **condition number** is $\kappa := \lambda_1/\lambda_N$, the largest variance in any direction over the smallest. It is 1 for a multiple of I and grows as the ellipse of Exhibit 2 flattens; a singular M has an infinite κ.

A 2×2 correlation matrix has eigenvalues $1 + \rho$ and $1 - \rho$, with eigenvectors $(1, 1)^\top/\sqrt{2}$ and $(1, -1)^\top/\sqrt{2}$, so

$$\kappa(C) = \frac{1 + |\rho|}{1 - |\rho|}.$$

This condition number measures LM7's warning about correlations near ±1. On LM8's sample, 0050 and 00631L have $\rho = 0.974365$, so $\kappa(C) = 77.0$: a correlation that looks merely high makes the variance in one direction 77 times that in another (Exhibit 5).

Exhibit 5: Condition numbers of correlation and covariance matrices

Assets	N	$\kappa(C)$	$\kappa(\hat{\Sigma})$
0050, 00679B (LM7's sample)	2	1.24	2.29
0050, 00631L (LM8's sample)	2	77.02	116.84
Semiconductor, FinancialInsurance	2	3.70	4.03
S3: the two and Food	3	5.88	7.04
S5: S3, Plastics, ShippingTransportation	5	8.52	12.12
U28	28	70.64	108.09
Electronic and its eight sub-industries	9	836.35	925.27
All 30 sector indices	30	2,265.72	2,410.67

Note: daily returns on the U28 sample, except LM7's 2,363 and LM8's 2,902 daily total returns in the first two rows. The covariance condition number is the same for daily and annualized units. It tends to grow with N; the second row is inflated by near-collinearity and the last two by overlap (Section 6).

EXAMPLE 4

Equal correlations

Let N assets have the same correlation $\rho > 0$, so $C = (1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top$. Find its eigenvalues and κ. Then use $\mathbf{u} = \mathbf{1}/\sqrt{N}$ in (11.4) to bound λ_1 of U28's correlation matrix, whose 756 off-diagonal entries average $\bar{\rho} = 0.50227$.

SOLUTION

Since $\mathbf{1}^\top\mathbf{1} = N$, $C\mathbf{1} = (1 + (N-1)\rho)\mathbf{1}$, so $1 + (N-1)\rho$ is an eigenvalue. Every $\mathbf{x}$ in the space of dimension $N-1$ orthogonal to $\mathbf{1}$ has $C\mathbf{x} = (1-\rho)\mathbf{x}$, so $1-\rho$ is an eigenvalue $N-1$ times, and $\kappa = (1 + (N-1)\rho)/(1-\rho)$.

For U28, $h(\mathbf{1}/\sqrt{N}) = \mathbf{1}^\top C\mathbf{1}/N = 1 + 27 \times 0.50227 = 14.561$, which by (11.4) is at most λ_1; indeed $\lambda_1 = 14.721$.

DEEP DIVE **Why LM7's gradient ascent contracts by (κ – 1)/(κ + 1)** optional · click to expand

In LM7's notation, $\mathbf{f}$ are exposures, g is the quadratic growth model with Hessian $-\Sigma$ and optimum $\mathbf{f}^*$, and α is the step. Since $\nabla g(\mathbf{f}) = -\Sigma(\mathbf{f} - \mathbf{f}^*)$, a step to $\mathbf{f} + \alpha\nabla g(\mathbf{f})$ turns the error $\mathbf{f} - \mathbf{f}^*$ into $(I - \alpha\Sigma)(\mathbf{f} - \mathbf{f}^*)$, which multiplies its component along $\mathbf{q}_k$ by $1 - \alpha\lambda_k$. With $\alpha = 2/(\lambda_1 + \lambda_N)$ the extreme factors are $\pm(\kappa - 1)/(\kappa + 1)$ and the others are smaller in size, so each step shrinks the distance to $\mathbf{f}^*$ at least by that factor. Example 2's eigenvalues give $(0.040789 - 0.017850)/(0.040789 + 0.017850) = 0.391$ at $\alpha = 34.1$, LM7's 0.39.

PITFALL **Say which matrix a condition number belongs to**

Rescaling the variables changes the eigenvalues, so $\kappa(C) \neq \kappa(\hat{\Sigma})$: for U28 they are 70.6 and 108.1. A daily and an annualized $\hat{\Sigma}$ share one κ, because multiplying a matrix by A multiplies every eigenvalue by A.

導讀 **方向上的變異數與條件數**

把 N 個報酬的一個固定組合看成單位向量（unit vector）$\mathbf{u}$（長度為 1，不是權重和為 1 的投資組合），它的樣本變異數（sample variance）就是 $\mathbf{u}^\top \hat{\Sigma} \mathbf{u}$。換到特徵向量（eigenvector）座標後，這個數是各特徵值（eigenvalue）的加權平均，權重加總為 1，所以一定落在最小與最大特徵值之間，最大值就在第一個特徵向量上。最大特徵值除以最小特徵值是條件數（condition number），代表橢圓有多扁。0050 與 00631L 的相關係數（correlation coefficient）0.974 看似只是「很高」，相關矩陣（correlation matrix）的條件數卻是 77：某個方向的變異數是另一個方向的 77 倍。相關矩陣與共變異數矩陣（covariance matrix）的條件數不同，引用時要說清楚是哪一個。

KNOWLEDGE CHECK 3

Find the condition number of a 2×2 correlation matrix with $\rho = 0.9$.

ANSWER

The eigenvalues are 1 ± 0.9, so the condition number is $1.9/0.1 = 19$.

5 PRINCIPAL COMPONENT ANALYSIS

- [ ] derive the first principal component as the unit vector of largest variance, with the eigenvalue as its Lagrange multiplier
- [ ] interpret explained shares, loadings and scores, and compare covariance with correlation PCA

Principal component analysis (PCA) asks which mix of the N returns varies most, then which varies most among the mixes uncorrelated with it, and so on. A mix is a vector $\mathbf{u}$; its centred return on date t is $\mathbf{u}^\top(\mathbf{R}_t - \bar{\mathbf{R}})$, with sample variance $\mathbf{u}^\top \hat{\Sigma} \mathbf{u}$.

The question needs a constraint, and portfolio weights, $\mathbf{1}^\top \mathbf{u} = 1$, do not work. The vector $(a, 1-a, 0, \ldots, 0)^\top$ satisfies them for every number a, and its variance grows like a^2 times the variance of

$R_{1,t} - R_{2,t}$. PCA fixes the length instead, $\mathbf{u}^\top \mathbf{u} = 1$, and compares directions.

The Lagrangian is $\mathcal{L} = \mathbf{u}^\top \hat{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$, with gradient $2\hat{\Sigma}\mathbf{u} - 2\lambda\mathbf{u}$ by LM10's (10.8). Setting it to zero gives $\hat{\Sigma}\mathbf{u} = \lambda\mathbf{u}$. So **every stationary point is a unit eigenvector, and its multiplier is the eigenvalue**, which is also the variance there: $\mathbf{u}^\top \hat{\Sigma} \mathbf{u} = \lambda\, \mathbf{u}^\top \mathbf{u} = \lambda$.

A continuous function on a closed and bounded set, such as the unit sphere, attains a largest value. This extreme value theorem is stated without proof; Strang and Herman (2016, §4.7, Theorem 4.18) give the case of two variables. So the maximum is the stationary point with the largest value:

$$\max_{\mathbf{u}^\top \mathbf{u} = 1} \mathbf{u}^\top \hat{\Sigma} \mathbf{u} = \lambda_1, \quad \text{attained at } \mathbf{u} = \mathbf{q}_1. \tag{11.5}$$

Maximizing again over unit vectors orthogonal to $\mathbf{q}_1$ gives $\mathbf{q}_2$, with variance λ_2, and so on. The directions $\mathbf{q}_k$ are the **principal components**, and their entries are **loadings**.

> **DEEP DIVE** **PCA proves the spectral theorem for every N** optional · click to expand
>
> Let M be real symmetric, with Lagrange multipliers λ and ζ. The maximum of $\mathbf{u}^\top M \mathbf{u}$ on the unit sphere exists, and the Lagrange condition makes it a real eigenpair $(\lambda_1, \mathbf{q}_1)$. Next maximize over unit vectors with $\mathbf{q}_1^\top \mathbf{u} = 0$. The gradient of $\mathbf{u}^\top M \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1) - \zeta\, \mathbf{q}_1^\top \mathbf{u}$ is $2M\mathbf{u} - 2\lambda\mathbf{u} - \zeta\mathbf{q}_1$, and its dot product with $\mathbf{q}_1$ is $-\zeta$, since $\mathbf{q}_1^\top M \mathbf{u} = \lambda_1 \mathbf{q}_1^\top \mathbf{u} = 0$. So $\zeta = 0$, and the maximizer is a real eigenpair $(\lambda_2, \mathbf{q}_2)$ with $\mathbf{q}_2 \perp \mathbf{q}_1$. Repeating N times gives (11.3), with no complex numbers and no characteristic polynomial.
>
> The construction proves (iii), and (i) follows. Let $M\mathbf{x} = \ell\mathbf{x}$ for a number ℓ and a vector $\mathbf{x} \neq \mathbf{0}$, both possibly complex. Expanding $\mathbf{x}$ in the orthonormal $\mathbf{q}_k$ (still valid, since $QQ^\top \mathbf{x} = \mathbf{x}$) gives $\sum_k (\lambda_k - \ell)(\mathbf{q}_k^\top \mathbf{x})\, \mathbf{q}_k = \mathbf{0}$, so every coefficient is zero. Some $\mathbf{q}_k^\top \mathbf{x}$ is not zero, so ℓ equals that λ_k and is real.

Shares. The eigenvalues add up to the trace, the sum of the N variances, so component k carries the **explained share**

$$\frac{\lambda_k}{\operatorname{tr} \hat{\Sigma}} = \frac{\lambda_k}{\lambda_1 + \cdots + \lambda_N} \qquad \left(= \frac{\lambda_k}{N} \text{ for } C \right) \tag{11.6}$$

of the total variance. A **scree plot** shows the shares against k (Exhibit 6).

EXAMPLE 5

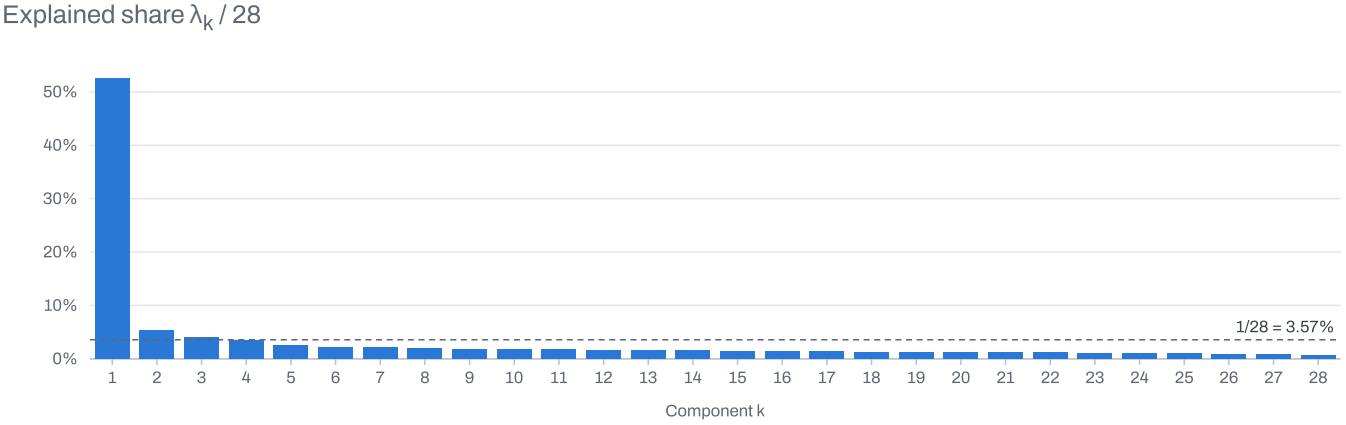
How many components does U28 need?

The correlation matrix of U28 has $\lambda_1 = 14.7214$ and $\lambda_2 = 1.4837$. Find their shares and the number of components needed to pass 80% of the total variance.

SOLUTION

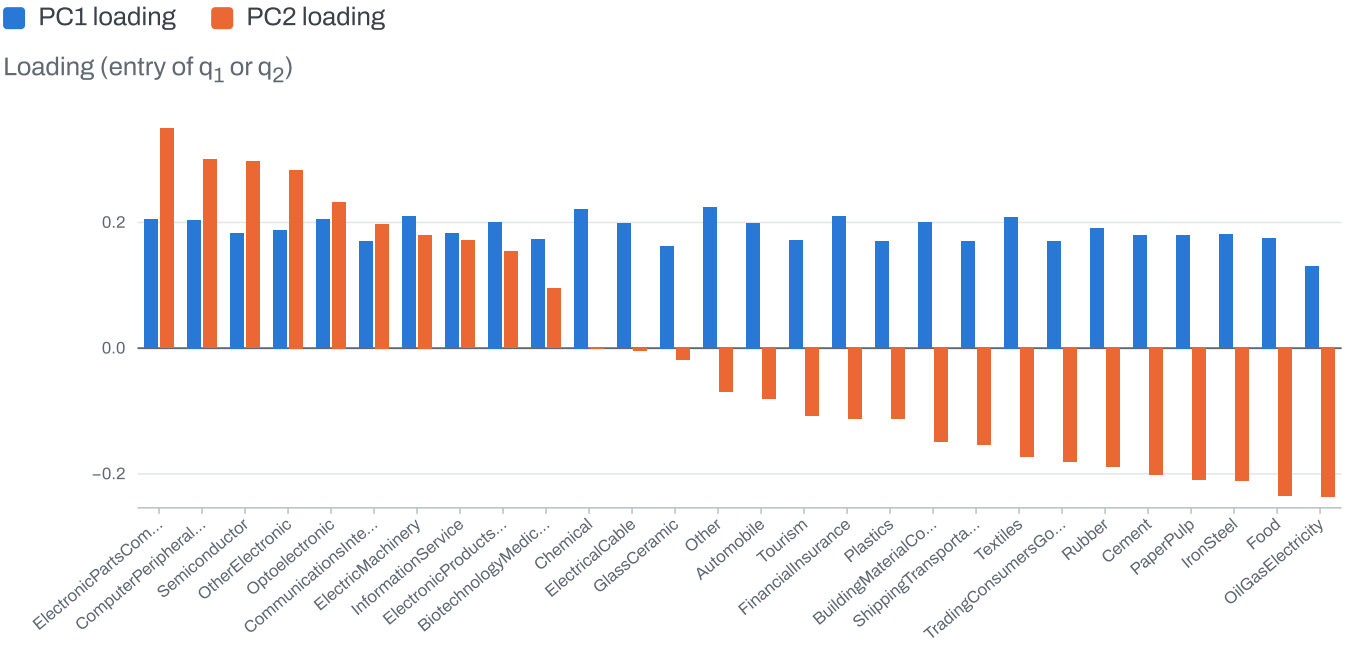
The trace is $N = 28$, so the shares are $14.7214/28 = 52.58\%$ and $1.4837/28 = 5.30\%$. The later shares decline slowly: 11 components carry 79.31% and 12 carry 80.93%, so 12 are needed.

Exhibit 6: Explained shares of U28's principal components



Note: $\lambda_k/28$ for the correlation matrix of U28. The dashed line, $1/28 = 3.57\%$, is every component's share if the 28 sectors were uncorrelated.

Exhibit 7: Loadings of U28's first two principal components



Note: entries of $\mathbf{q}_1$ and $\mathbf{q}_2$ of U28's correlation matrix, sectors sorted by their PC2 loading. The PC1 loadings run from 0.130 (OilGasElectricity) to 0.224 (Other), around the equal loading $1/\sqrt{28} = 0.189$.

Scores. The score of component k on date t in correlation PCA is

$$\xi_{k,t} = \mathbf{q}_k^\top \operatorname{diag}(\mathbf{s})^{-1}(\mathbf{R}_t - \bar{\mathbf{R}}),$$

the day's standardized returns read along $\mathbf{q}_k$. Their covariance matrix is C, so $\operatorname{Var}\xi_k = \mathbf{q}_k^\top C\mathbf{q}_k = \lambda_k$ (divisor T), and two different scores, ξ_k and $\xi_{k'}$, have covariance $\mathbf{q}_{k'}^\top C\mathbf{q}_k = \lambda_k \mathbf{q}_{k'}^\top \mathbf{q}_k = 0$. PCA turns N correlated series into N uncorrelated ones, ranked by variance.

Covariance or correlation. Covariance PCA decomposes $\hat{\Sigma}$, correlation PCA decomposes C. New units for one variable rescale its row and column of $\hat{\Sigma}$ but leave C alone. On U28 the two analyses agree on the first component but not on the second (Exhibit 8).

Exhibit 8: Covariance and correlation PCA of U28

	Covariance PCA ($\hat{\Sigma}$)	Correlation PCA (C)
Share of PC1	51.70%	52.58%
Share of PC2	5.88%	5.30%
Largest PC2 loading	0.667, GlassCeramic	0.350, ElectronicPartsComponents

Note: full sample. GlassCeramic is the most volatile sector, at 35.31% a year. A covariance share is a share of the sum of the 28 variances, not of a portfolio's variance. The two PC1 vectors have $|\cos\theta| = 0.9791$.

EXAMPLE 6

Restating one sector in percent

Multiply Semiconductor's returns by 100 and keep FinancialInsurance in decimals. What happens to the first component of covariance PCA and of correlation PCA?

SOLUTION

Semiconductor's variance grows by a factor of 10^4 and the covariance by 100. Covariance PC1 becomes $(1.0000,\ 0.0046)^\top$ with a share of 99.996%: Semiconductor alone. Correlations do not depend on units, so correlation PCA keeps $\mathbf{q}_1 = (1,1)^\top/\sqrt{2}$ with eigenvalue $1 + 0.5749 = 1.5749$, and $\mathbf{q}_2 = (1,-1)^\top/\sqrt{2}$ with 0.4251.

RESULT **The case, first question**

U28 has one dominant direction, with 52.6% of the standardized variance and every loading positive (Exhibit 7): the sectors mostly move together. Beyond it no direction stands out, and 12 components are needed to pass 80%.

PITFALL **Naming a component**

PCA lets the data choose the directions, and nothing makes PC1 "the market" or PC2 "electronics". An economic factor model names its factors in advance and estimates exposures by regression (LM23). Some texts call $\mathbf{q}_k$ an eigen-portfolio; here it is a direction in the data, since its entries need not sum to 1 and the indices are price indices.

導讀 **主成分分析：讓資料自己選方向**

主成分分析（principal component analysis）找的是變異數（variance）最大的方向：在長度為 1 的向量（vector）中求 $\mathbf{u}^\top \hat{\Sigma} \mathbf{u}$ 的最大值。拉格朗日乘數（Lagrange multiplier）法的駐點（stationary point）一定是特徵向量（eigenvector），乘數就是特徵值（eigenvalue）。限制必須是長度 $\mathbf{u}^\top \mathbf{u} = 1$，不能是權重和為 1，否則變異數可以無限大。28 個產業中，第一主成分占標準化變異數的 52.6%，負荷量（loading）全部為正，表示產業大多同漲同跌；但要 12 個主成分才超過 80%。共變異數矩陣（covariance matrix）的主成分會被波動最大的產業主導，也會隨單位改變，相關矩陣（correlation matrix）則不受單位影響。

KNOWLEDGE CHECK 4

You restate FinancialInsurance's returns in basis points. Which analysis stays the same, covariance PCA or correlation PCA?

ANSWER

Correlation PCA, because correlations do not depend on units. Covariance PCA would swing toward FinancialInsurance, as Example 6 shows for Semiconductor.

6 OVERLAPPING INDICES AND NEAR-ZERO EIGENVALUES

- ☐ interpret explained shares, loadings and scores, and compare covariance with correlation PCA
- ☐ calculate the condition number $\kappa = \lambda_1 / \lambda_N$ and explain why solving a system with $\hat{\Sigma}$ amplifies estimation error

If λ_N is close to zero, the mix $\mathbf{q}_N^\top (\mathbf{R}_t - \bar{\mathbf{R}})$ has a mean square of only λ_N and stays near zero, so the series obey a near-exact linear relation. When $\lambda_N = 0$ the relation is exact and $\hat{\Sigma}$ is singular, the perfect collinearity of LM9. How small counts as zero in floating point is a question of numerical rank (LM12).

EXAMPLE 7

An aggregate and its members

Electronic is a market-value-weighted price index of the stocks in its eight electronics sub-industries (TWSE 2007, 2023). Read the smallest eigenpair of the correlation matrix of Electronic and the eight.

SOLUTION

The smallest eigenvalue is $\lambda_9 = 0.0073$, against $\lambda_1 = 6.104$, so $\kappa(C) = 836$. Its eigenvector has entries 0.828 for Electronic and −0.529 for Semiconductor, and the other seven are at most 0.104 in size. With almost no variance along $\mathbf{q}_9$, the standardized Electronic return stays close to 0.64 times the standardized Semiconductor return, plus smaller multiples of the others.

Exhibit 9: Near-zero eigenvalues of overlapping indices

Correlation matrix	Smallest eigenvalue	Largest entries of its eigenvector
Electronic and its eight sub-industries	0.0073	Electronic 0.828, Semiconductor −0.529
All 30 sector indices	0.00711	Electronic 0.826, Semiconductor −0.530
All 30, second smallest	0.01517	ChemicalBiotechnologyMedicalCare 0.771, Chemical −0.473, BiotechnologyMedicalCare −0.425
U28	0.2084	no aggregate

Note: daily returns on the U28 sample. Each aggregate adds one near-zero eigenvalue.

PITFALL **An aggregate index beside its members**

Putting Electronic, or TAIEX, beside its members adds a direction of almost no variance. The condition number then reflects how the index is built, not how markets move: $\kappa(C)$ jumps from 70.64 for U28 to 2,265.72 for all 30 indices (Exhibit 5, Exhibit 9). The book's universe therefore holds only the 28 leaves.

KNOWLEDGE CHECK 5

In Example 7, what would λ_9 be if Electronic's return were a fixed combination of the eight sub-industries' returns on every day? Why is the real value not that?

ANSWER

Zero: a combination of the nine series would vanish on every date, and C would be singular. Electronic weights each sub-industry by its market value at the previous close, and those weights drift with prices, so the relation is only nearly exact.

7 EIGENVALUE GAPS AND THE STABILITY OF COMPONENTS

- ☐ explain the sign convention for eigenvectors and why near-equal eigenvalues make their eigenvectors unstable
- ☐ compare principal components across calendar years and halves of a sample, and label the comparison a stability check

How far does an eigenvector turn when the matrix changes a little? Rotate the axes of a symmetric 2×2 matrix by an angle θ. The unit vectors $\mathbf{q}_1 = (\cos\theta, \sin\theta)^\top$ and $\mathbf{q}_2 = (-\sin\theta, \cos\theta)^\top$ are its eigenvectors exactly when $M\mathbf{q}_1$ has no component along $\mathbf{q}_2$:

$$\begin{aligned} \mathbf{q}_2^\top M \mathbf{q}_1 &= m_{12}\cos 2\theta \\ &\quad - \tfrac{1}{2}(m_{11} - m_{22})\sin 2\theta = 0. \end{aligned}$$

Start from a diagonal matrix with neighbouring eigenvalues $\lambda_k > \lambda_{k+1}$, and add a small ε to both off-diagonal entries. The eigenvectors turn by

$$\theta = \tfrac{1}{2}\arctan\frac{2\varepsilon}{\lambda_k - \lambda_{k+1}} \approx \frac{\varepsilon}{\lambda_k - \lambda_{k+1}}, \qquad \text{from} \quad \tan 2\theta = \frac{2m_{12}}{m_{11} - m_{22}}. \tag{11.7}$$

The arctangent is exact; the approximation drops terms of third order in ε over the gap. **The rotation is the perturbation divided by the gap**, so a gap no larger than ε can turn the eigenvectors by tens of degrees, up to 45°. Equal correlations are the limit, with every gap below λ_1 zero (Example 4).

EXAMPLE 8

How far noise can turn a component

U28's correlation matrix has $\lambda_2 = 1.48370$, $\lambda_3 = 1.14694$, $\lambda_7 = 0.58028$ and $\lambda_8 = 0.54445$. Let noise add $\varepsilon = 0.05$ to the entry linking two neighbouring eigenvectors. Find the rotation by (11.7),

exactly and to first order.

SOLUTION

For components 2 and 3 the gap is 0.33676, and ε over the gap is 0.1485. The exact angle is $\frac{1}{2}\arctan(0.1/0.33676) = 8.27^\circ$; to first order it is 0.1485 radians, 8.51°.

For components 7 and 8 the gap is 0.03583, smaller than ε. The exact angle is 35.1°, but the first-order formula gives 1.3955 radians, 80.0°. It fails once ε is not small next to the gap, and components 7 and 8 are barely determined.

Calendar years. Each year from 2009 to 2026 (2026 to 8 October, 185 returns) gives its own correlation matrix. December 2008's 22 returns enter only the full sample and the halves. The first component's share moves from 34.5% (2017) to 66.6% (2020) (Exhibit 10). Its direction hardly moves: its cosine with the full sample's $\mathbf{q}_1$ is at least 0.98 every year, and lowest in 2026, at 0.981 (Exhibit 11).

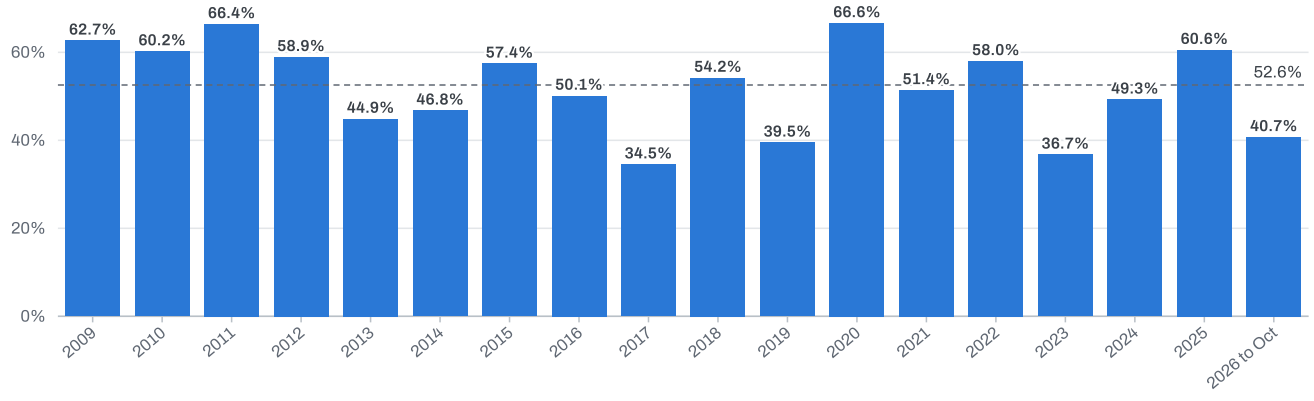
The second component's cosine runs from 0.119 (2016) to 0.946 (2025). When λ_2 and λ_3 are close, one period's $\mathbf{q}_2$ can be a mix of another period's $\mathbf{q}_2$ and $\mathbf{q}_3$. With the first half as the reference, the second half's $\mathbf{q}_2$ has the **plane measure**

$$\sqrt{\left(\mathbf{q}_2^{(1)\top}\mathbf{q}_2^{(2)}\right)^2 + \left(\mathbf{q}_3^{(1)\top}\mathbf{q}_2^{(2)}\right)^2}.$$

Since $\mathbf{q}_2^{(1)}$ and $\mathbf{q}_3^{(1)}$ are orthonormal, this measure is the cosine of the angle between $\mathbf{q}_2^{(2)}$ and their plane. The measure is one-sided: it shows where the compared period's $\mathbf{q}_2$ went, not that the plane holds. For a year's $\mathbf{q}_2$ against the full sample's plane, it runs from 0.690 (2020) to 0.959 (2024).

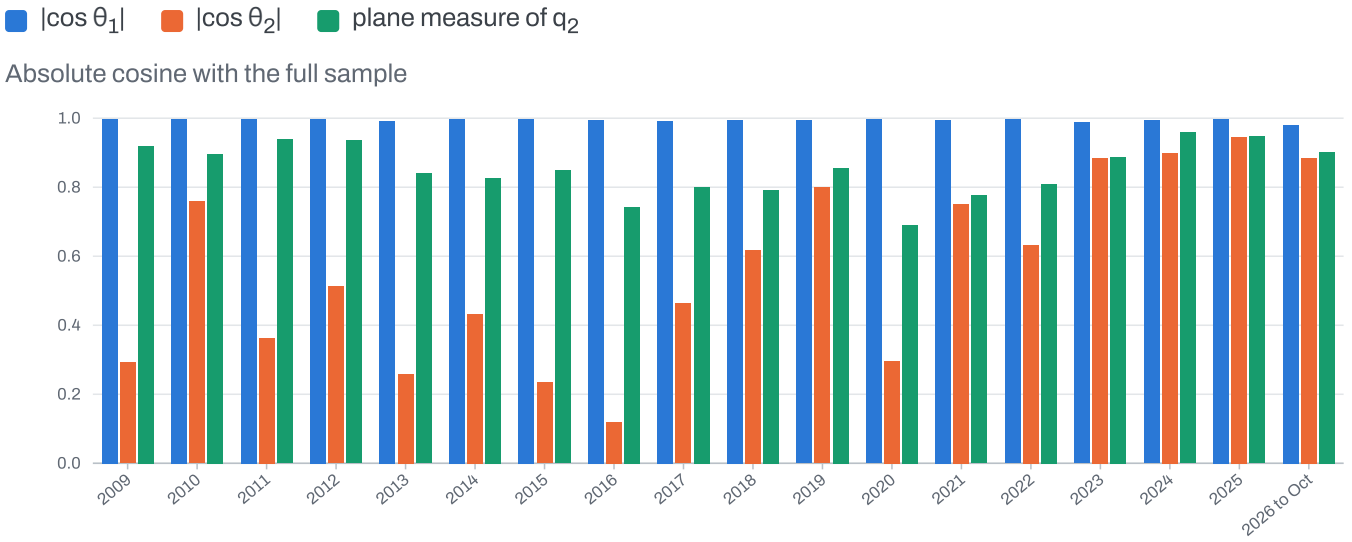
Exhibit 10: Share of the first principal component by calendar year

First component's share λ_1 / 28



Note: $\lambda_1/28$ of each calendar year's correlation matrix of U28; 2026 runs to 8 October (185 returns). The dashed line is the full-sample share, 52.6%.

Exhibit 11: Each year's components against the full sample's



Note: $|\cos\theta|$ of each year's $\mathbf{q}_1$ and $\mathbf{q}_2$ with the full sample's, and the plane measure of the year's $\mathbf{q}_2$ against the full sample's $\mathbf{q}_2$ and $\mathbf{q}_3$. The plane measure shows where that $\mathbf{q}_2$ lies, not that the plane holds. Correlation matrices of U28; 2026 runs to 8 October.

Two halves. The halves hold 2,191 returns each (Exhibit 12). The first components agree almost perfectly, $|\cos\theta_1| = 0.997$; the second do not, $|\cos\theta_2| = 0.297$. PC2's largest loading is OilGasElectricity (0.572) in the first half and ElectronicPartsComponents (0.341) in the second, with a signed cosine of −0.297. The plane measure is $\sqrt{0.2966^2 + 0.8787^2} = 0.9274$, an angle of 22°: the second half's $\mathbf{q}_2$ is mostly the first half's $\mathbf{q}_3$ (cosine 0.879).

The reverse measure, the first half's $\mathbf{q}_2$ against the second half's plane, is only $\sqrt{0.2966^2 + 0.4196^2} = 0.514$. That vector is mostly the second half's $\mathbf{q}_4$ (cosine 0.776): beyond the first, the components changed places. The plane does not hold either. The plane measure of the second half's $\mathbf{q}_3$ is only 0.437, an angle of 64° from the first half's plane. In that half, λ_3 and λ_4 differ by only 0.069, so (11.7) leaves $\mathbf{q}_3$ poorly determined.

Exhibit 12: Principal components of U28 in the two halves of the sample

k	$\lambda_k^{(1)}$	$\lambda_k^{(2)}$	$\lvert\mathbf{q}_k^{(1)\top}\mathbf{q}_k^{(2)}\rvert$
1	16.116	13.787	0.997
2	1.262	1.848	0.297
3	1.069	1.171	0.121
4	0.666	1.102	0.043
5	0.619	0.740	0.428

Note: correlation matrices of the first 2,191 returns (2 December 2008 to 6 October 2017) and the last 2,191 (11 October 2017 to 8 October 2026). PC1's share is 57.6% and 49.2%.

RESULT **The case, second question**

The first direction holds over time, with cosines of at least 0.98 every year and 0.997 across the halves. Only its share moves, between 34.5% and 66.6%. The second direction does not hold (0.297): the first half's third direction ranks second in the second half (0.879), and its second ranks fourth (0.776). Beyond the first direction the components change places, so only the dominant direction is worth describing.

PITFALL **A stability check is not an out-of-sample test**

Comparing the halves shows how much the estimates move; it tests nothing. An out-of-sample test fits on earlier data and judges on later data that the fit never used (Research hygiene, LM34). An eigenvalue estimated from 250 returns also has sampling error of its own, and this module attaches no confidence intervals (LM20, LM22).

導讀 **特徵值間距決定主成分穩不穩**

矩陣受到一點擾動（perturbation）ε 時，特徵向量（eigenvector）轉動的角度大約是 ε 除以兩個特徵值（eigenvalue）的間距；間距很小時，一點雜訊就能讓方向轉動幾十度。28 個產業中，第一個特徵值遠大於其他，所以第一主成分（principal component）在每一年、前後兩半都幾乎不動。第二個以後的特徵值小而接近：第二主成分在兩半之間的餘弦（cosine）只有 0.297，後半段的第二主成分大致是前半段的第三主成分（0.879），前半段的第二主成分則落到第四（0.776），第二、第三主成分張成的平面也不穩定。平面度量（plane measure）只看單邊，不能證明平面穩定。比較兩個期間的特徵向量要看餘弦的絕對值，不要逐一比較元素；這是穩定性檢查（stability check），不是樣本外檢定（out-of-sample test）。

KNOWLEDGE CHECK 6

Why is U28's first component stable across the halves while its fourth is not?

ANSWER

In the full sample λ_1 lies 13.24 above λ_2, so by (11.7) noise barely turns $\mathbf{q}_1$. Yet λ_4 lies only 0.193 below λ_3 and 0.256 above λ_5, so the same noise can turn $\mathbf{q}_4$ by more than ten degrees: by 13.7° against $\mathbf{q}_3$ for $\varepsilon = 0.05$. The halves agree: $|\cos\theta_1| = 0.997$, $|\cos\theta_4| = 0.043$.

8 THE INVERSE OF $\hat{\Sigma}$ AND ERROR AMPLIFICATION

- [ ] calculate the condition number $\kappa = \lambda_1/\lambda_N$ and explain why solving a system with $\hat{\Sigma}$ amplifies estimation error

Rescaled to sum to 1, the solution of $\hat{\Sigma}\mathbf{x} = \mathbf{1}$ is the minimum-variance portfolio of LM28; here it is only a linear system. When every $\lambda_k > 0$, the inverse of $Q\Lambda Q^\top$ is $Q\Lambda^{-1}Q^\top$, because their product is $QQ^\top = I$. Written as a sum,

$$\hat{\Sigma}^{-1}\mathbf{b} = \sum_{k=1}^{N} \frac{\mathbf{q}_k^\top \mathbf{b}}{\lambda_k}\,\mathbf{q}_k. \tag{11.8}$$

Each direction enters with the weight $1/\lambda_k$. **The smallest eigenvalues get the largest weights.** In U28 they also lie close together, so by (11.7) their eigenvectors are poorly determined. Between the halves' covariance matrices, 12 of the 14 eigenvectors of the smallest eigenvalues have $|\cos\theta|$ below 0.25.

The spectral norm. The **spectral norm** $\|M\|_2$ is the largest length of $M\mathbf{u}$ over unit vectors $\mathbf{u}$. For symmetric M, $\|M\mathbf{u}\|^2 = \mathbf{u}^\top M^2 \mathbf{u}$ and $M^2 = Q\Lambda^2 Q^\top$, so by (11.4) $\|M\|_2$ is the largest $|\lambda_k|$. A positive definite M thus has $\|M\|_2 = \lambda_1$, $\|M^{-1}\|_2 = 1/\lambda_N$ and $\kappa = \|M\|_2\|M^{-1}\|_2$.

PROPOSITION **How far a solution can move**

Let M be positive definite with $M\mathbf{x} = \mathbf{b}$, and let a change ΔM move the solution to $\mathbf{x} + \Delta\mathbf{x}$, so that $(M + \Delta M)(\mathbf{x} + \Delta\mathbf{x}) = \mathbf{b}$. Then

$$\frac{\|\Delta\mathbf{x}\|}{\|\mathbf{x} + \Delta\mathbf{x}\|} \le \kappa\,\frac{\|\Delta M\|_2}{\|M\|_2}. \tag{11.9}$$

Proof. Subtracting $M\mathbf{x} = \mathbf{b}$ leaves $M\Delta\mathbf{x} = -\Delta M(\mathbf{x} + \Delta\mathbf{x})$, so $\Delta\mathbf{x} = -M^{-1}\Delta M(\mathbf{x} + \Delta\mathbf{x})$. Since $\|K\mathbf{y}\| \le \|K\|_2\|\mathbf{y}\|$ for any matrix K and vector $\mathbf{y}$, $\|\Delta\mathbf{x}\| \le \|M^{-1}\|_2\|\Delta M\|_2\|\mathbf{x} + \Delta\mathbf{x}\|$. Finally $\|M^{-1}\|_2 = 1/\lambda_N = \kappa/\|M\|_2$. ■

The change ΔM is usually indefinite, so $\|\Delta M\|_2$ is its largest eigenvalue in absolute value, not its λ_1. The bound is a worst case, nearly reached only when the change pushes the solution along the eigenvector of λ_N.

EXAMPLE 9

A bound nearly reached

Let C be the correlation matrix of two assets with $\rho = 0.95$, and solve $C\mathbf{x} = \mathbf{e}_1$. Then raise ρ to 0.96, and compare the relative change of the solution with the bound (11.9) for $M = C$.

SOLUTION

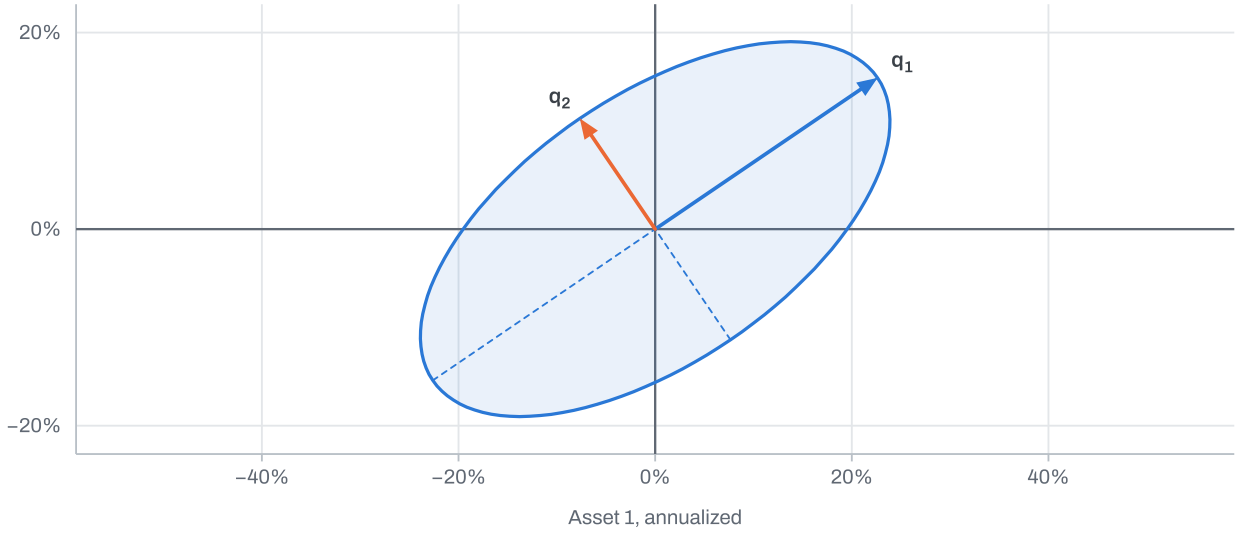
By LM9's (9.5), $\mathbf{x} = (1, -\rho)^\top/(1-\rho^2)$. At $\rho = 0.95$, $\mathbf{x} = (10.256,\ -9.744)^\top$; at 0.96, $\mathbf{x} + \Delta\mathbf{x} = (12.755,\ -12.245)^\top$. So $\Delta\mathbf{x} = (2.499,\ -2.501)^\top$, and the relative change is 0.19996.

Here $\kappa = 1.95/0.05 = 39$, and ΔM has 0.01 off the diagonal, with eigenvalues ±0.01, so $\|\Delta M\|_2/\|M\|_2 = 0.01/1.95 = 0.005128$. The bound $39 \times 0.005128 = 0.2000$ is almost reached, because $\Delta\mathbf{x}$ lies along $(1, -1)^\top/\sqrt{2}$, the eigenvector of the small eigenvalue $1-\rho$.

The widget repeats the experiment of Example 9 for any two assets.

Two assets: eigenvectors, condition number and a moving solution

INTERACTIVE



Eigenvalues $\lambda_1 \cdot \lambda_2$	Share of the first component	Condition number $\kappa = \lambda_1/\lambda_2$
0.0748 · 0.0186	**80.1%**	**4.03**
Angle of q_1 from axis 1	**Solution x of Mx = b**	**After ρ + 0.01**
34.2°	**(26.19, −18.86)**	**(26.65, −19.52)**
Relative change of x	**Bound (11.9)**	
0.0245	**0.0245**	

The ellipse is the set of points x with $x^T M^{-1} x = 1$; its half-axes, $\sqrt{\lambda_1}$ and $\sqrt{\lambda_2}$, lie along the eigenvectors q_1 and q_2 (signed by the book's rule). The last rows solve Mx = b, then raise ρ by 0.01 and solve again. The relative change of the solution never exceeds the bound (11.9): κ times the change of the matrix relative to its spectral norm. Initial state: Semiconductor (asset 1) and FinancialInsurance (asset 2), 2008–2026. With equal volatilities and b = 1 = (1, 1), the vector 1 is an eigenvector. For ρ > 0 it is q_1, and the solution barely moves (relative change 0.01/(1 + ρ)). For ρ < 0 it is q_2, the direction of the small eigenvalue, and the change reaches the bound.

EXAMPLE 10

Where the solution of $\hat{\Sigma}\mathbf{x} = \mathbf{1}$ comes from

Solve $\hat{\Sigma}^{(1)}\mathbf{x}^{(1)} = \mathbf{1}$ and $\hat{\Sigma}^{(2)}\mathbf{x}^{(2)} = \mathbf{1}$ for U28 on the two halves. Split each $\|\mathbf{x}\|^2$ by component with (11.8), and compare the realized change with the bound (11.9).

SOLUTION

The $\mathbf{q}_k$ are orthonormal, so by (11.8) component k contributes $(\mathbf{q}_k^\top \mathbf{1}/\lambda_k)^2/\|\mathbf{x}\|^2$ (Exhibit 14). PC1, the most stable direction, contributes 1.8% in the first half and 1.0% in the second; the 14 smallest components contribute 72.0% and 80.5%.

The matrix moved by $\|\hat{\Sigma}^{(2)} - \hat{\Sigma}^{(1)}\|_2/\|\hat{\Sigma}^{(1)}\|_2 = 0.24536$, with $\kappa(\hat{\Sigma}^{(1)}) = 126.05$, so (11.9) allows up to $126.05 \times 0.24536 = 30.93$. The realized change, 0.729 (Exhibit 13), is far below that worst case.

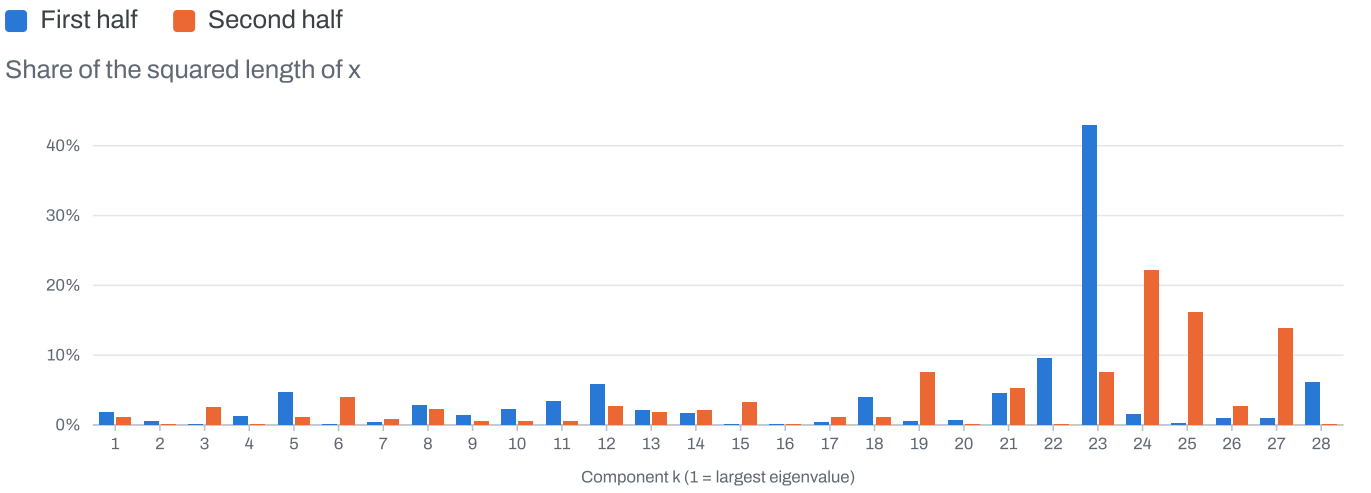
Exhibit 13: Solutions of $\hat{\Sigma}\mathbf{x} = \mathbf{1}$ in the two halves

S5 sector	$\mathbf{x}^{(1)}$	$\mathbf{x}^{(2)}$
Semiconductor	2,290.29	−23.72
FinancialInsurance	−1,013.59	5,609.00
Food	3,271.34	9,584.83
Plastics	2,591.12	−281.56
ShippingTransportation	2,279.27	49.08

Summary	S5	U28
Relative change $\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|/\|\mathbf{x}^{(2)}\|$	0.910	0.729
Cosine of $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$	0.419	0.690
$\kappa(\hat{\Sigma}^{(1)})$	10.863	126.05
$\|\hat{\Sigma}^{(2)} - \hat{\Sigma}^{(1)}\|_2/\|\hat{\Sigma}^{(1)}\|_2$	0.44235	0.24536
Bound (11.9)	4.805	30.93

Note: daily covariance matrices of the two halves of Exhibit 12; U28's entries for Semiconductor and FinancialInsurance are in Exhibit 1.

Exhibit 14: Share of the squared length of $\hat{\Sigma}^{-1}1$ from each component



Note: $(\mathbf{q}_k^\top \mathbf{1}/\lambda_k)^2/\|\mathbf{x}\|^2$ for each half's daily $\hat{\Sigma}$ of U28, from the largest eigenvalue ($k = 1$) to the smallest ($k = 28$). Components 15 to 28 contribute 72.0% and 80.5%.

RESULT **The case, third question**

Between the halves U28's covariance matrix moved by 24.5% in the spectral norm, and the solution of $\hat{\Sigma}\mathbf{x} = \mathbf{1}$ by 0.729 of the second half's solution length. The inverse weights each direction by $1/\lambda_k$, so most of the solution comes from the smallest, poorly determined components. Pulling the small sample eigenvalues up lowers κ; covariance shrinkage, the subject of LM30, does this.

PITFALL **What a condition number does not say**

A large κ does not mean that a computed solution is wrong; it says how far the solution could move for a given change in the matrix. Nor does it rank realized changes. S5 has $\kappa(\hat{\Sigma}^{(1)}) = 10.9$ and its solution moved by 0.910, while U28 has $\kappa(\hat{\Sigma}^{(1)}) = 126$ and moved by 0.729 (Exhibit 13).

導讀 **反矩陣為什麼會放大誤差**

由譜分解（spectral decomposition），反矩陣（inverse matrix）$\hat{\Sigma}^{-1}$ 把 $\mathbf{b}$ 在每個特徵向量（eigenvector）方向上的分量乘上 $1/\lambda_k$，特徵值（eigenvalue）越小的方向權重越大。偏偏最小的那些特徵值彼此接近、方向很不穩定，所以解 $\hat{\Sigma}\mathbf{x} = \mathbf{1}$ 時，矩陣只動了 24.5%，解的相對變動卻是 0.729。條件數（condition number）κ 給的是最壞情況：相對誤差最多放大 κ 倍，28 個產業的上界是 30.93，實際只有 0.729。條件數大不代表算錯，也不能用來排序實際變動的大小。把小特徵值往上拉可以降低條件數，這是 LM30 的主題。

KNOWLEDGE CHECK 7

Where does most of $\hat{\Sigma}^{-1}\mathbf{1}$ come from, the first component or the last ones? Why?

ANSWER

From the last ones. By (11.8) component k enters with the weight $1/\lambda_k$, so small eigenvalues dominate unless $\mathbf{q}_k^\top \mathbf{1}$ is tiny. For U28 the 14 smallest components give 72.0% and 80.5% of $\|\mathbf{x}\|^2$.

9 EIGEN-DECOMPOSITION IN PYTHON

- ☐ calculate the eigenvalues and eigenvectors of a 2×2 symmetric matrix by hand, and of an $N \times N$ one with `eigh`
- ☐ explain the sign convention for eigenvectors and why near-equal eigenvalues make their eigenvectors unstable

For symmetric matrices, NumPy's `eigh` returns the eigenvalues in ascending order (NumPy reference), so reverse them, and signs the eigenvectors arbitrarily, so apply the sign rule. It uses an iterative method, not the characteristic polynomial (stated without proof; Golub and Van Loan 2013, ch. 8).

The same components also come from the singular value decomposition of $\tilde{X}/\sqrt{T}$ (LM12, (12.5); Strang 2016, §7.3), the route software prefers. The squared singular values are the eigenvalues of $\hat{\Sigma}$. For correlation PCA, decompose the standardized returns $\tilde{X}\,\mathrm{diag}(\mathbf{s})^{-1}/\sqrt{T}$ instead.

`np.linalg.cond` uses the spectral norm, so it equals λ_1/λ_N only for symmetric positive definite input. Solve systems with `solve`, never with `inv` (LM9).

PYTHON

```python
import numpy as np

def eig_desc(M):
    """Eigenpairs of a symmetric M, largest first, signed by the book's rule."""
    lam, Q = np.linalg.eigh(M)              # ascending order
    lam, Q = lam[::-1], Q[:, ::-1].copy()   # largest first
    for k in range(Q.shape[1]):
        a = np.abs(Q[:, k])
        i = np.flatnonzero(a >= a.max() * (1 - 1e-12))[0]   # first of ties
        if Q[i, k] < 0:
            Q[:, k] = -Q[:, k]
    return lam, Q

# X: the T x N array of daily returns (lab, Part 1)
S = np.cov(X, rowvar=False, ddof=0)
lam, Q = eig_desc(np.corrcoef(X, rowvar=False))
print(lam[:2] / lam.sum())                  # explained shares (11.6)
lamS, QS = eig_desc(S)
print(np.linalg.cond(S), lamS[0] / lamS[-1])  # equal: S is SPD
x = np.linalg.solve(S, np.ones(X.shape[1]))   # never inv(S) @ ones
```

IN PYTHON **Companion lab**

The Polars-first notebook `LM11_lab.ipynb` repairs the sector closes and rebuilds every exhibit and computational example. It solves the computational practice problems and ends with an extension, the first component's share over rolling windows of 250 returns.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 8

Which eigenpair does `np.linalg.eigh` list first, and what must you do before you read $\mathbf{q}_1$ from its output?

ANSWER

The smallest eigenvalue and its eigenvector. Reverse the order so that λ_1 comes first, and apply the sign rule before you compare or print eigenvectors.

SUMMARY

- An eigenvector is a direction that a matrix only stretches, $M\mathbf{x} = \lambda_k \mathbf{x}$; the eigenvalues solve $\det(M - \lambda_k I) = 0$ (11.1)–(11.2) and add up to the trace.
- A real symmetric matrix has real eigenvalues and orthonormal eigenvectors (11.3); it is positive semidefinite exactly when every $\lambda_k \geq 0$.
- Every unit direction has a variance between λ_N and λ_1 (11.4). The condition number $\kappa = \lambda_1/\lambda_N$ of the correlation matrix is 70.6 for U28 and 77.0 for 0050 and 00631L.
- The first principal component maximizes variance over unit vectors, with the eigenvalue as multiplier (11.5). In U28 it carries 52.6% of the standardized variance; 12 components reach 80%.
- Eigenvectors turn by about the perturbation divided by the gap (11.7). U28's first component is stable (cosine at least 0.98 every year); beyond it, the components of the two halves change places.
- The inverse weights direction k by $1/\lambda_k$ (11.8), so solutions come mostly from small, poorly determined directions. The bound (11.9) is a worst case: 30.93 for U28's halves, against a realized 0.729.

PRACTICE PROBLEMS

1. Which matrices are guaranteed to have real eigenvalues and orthogonal eigenvectors for different eigenvalues?

   A. Every matrix with positive entries

   B. Every real symmetric matrix

   C. Every invertible matrix

2. The correlation matrix of U28 has $\lambda_1 = 14.721$. The explained share of the first principal component is closest to:

   A. 14.7%

   B. 3.6%

   C. 52.6%

3. An analyst restates one sector's returns in percent and leaves the others in decimals. What changes?

   A. The loadings of covariance PCA

   B. The eigenvalues of correlation PCA

   C. The loadings of correlation PCA

4. Across the two halves, U28's second components have $|\cos\theta_2| = 0.297$. The plane measure of the second half's $\mathbf{q}_2$ against the first half's $\mathbf{q}_2$ and $\mathbf{q}_3$ is 0.927. The reverse measure, the first half's $\mathbf{q}_2$ against the second half's plane, is 0.514. Which reading is best?

A. Beyond the first, the components changed places.

B. The plane of the second and third directions holds.

C. The sign rule failed.

5. The daily covariance matrix of Plastics and Food on the U28 sample is

$$10^{4}\hat{\Sigma} = \begin{pmatrix} 2.5228 & 0.7180 \\ 0.7180 & 1.1550 \end{pmatrix}.$$

From $\mathbf{q}_1$ these printed entries, find its eigenvalues to four decimals, , the first explained share and the condition number.

6. (a) Find the eigenvalues, first share and κ of an equicorrelation matrix with $N = 5$ and $\rho = 0.5$. (b) S5's correlation matrix has $\lambda_1 = 2.8633$, $\lambda_5 = 0.3362$ and $\bar{\rho} = 0.462$. Compare with (a), and check (11.4) at $\mathbf{u} = \mathbf{1}/\sqrt{5}$.

7. Build the symmetric 2×2 matrix with eigenvalues 4 and 1 whose first eigenvector makes an angle of 30° with the first axis. Check its trace and determinant.

8. For the matrix with rows $(2, \varepsilon)$ and $(\varepsilon, 1)$, find the angle of $\mathbf{q}_1$ with the first axis by (11.7), exactly and to first order, for $\varepsilon = 0.1$ and $\varepsilon = 0.5$.

9. (a) Which correlation $\rho > 0$ gives a 2×2 correlation matrix the condition number 100? (b) On LM8's sample, 0050 and 00631L have $\rho = 0.974365$. Find $\kappa(C)$.

10. For S5's halves, $\kappa(\hat{\Sigma}^{(1)}) = 10.863$, the matrix changes by 0.44235 of its spectral norm, and the solution of $\hat{\Sigma}\mathbf{x} = \mathbf{1}$ changes by 0.910. Compute the bound (11.9) and compare with U28 (Exhibit 13). Does U28's larger κ mean that its solution moved more?

11. *(Python)* For S5 on the U28 sample, compute the correlation matrix, its eigenvalues and signed loadings with `eig_desc`, and the scores. Check that $\operatorname{Var} \xi_k = \lambda_k$ (divisor T) and that different scores are uncorrelated.

12. *(Python)* For each calendar year from 2009 to 2026, compare U28's correlation PCA with the full sample's. Compute $|\cos\theta|$ for $\mathbf{q}_1$ and $\mathbf{q}_2$, and the plane measure of the year's $\mathbf{q}_2$ against the full sample's $\mathbf{q}_2$ and $\mathbf{q}_3$. Which year has the smallest gap $\lambda_2 - \lambda_3$?

SOLUTIONS

1. **B** is correct, by the spectral theorem. A matrix with positive entries need not be symmetric: the one with rows $(1, 2)$ and $(3, 4)$ has two real eigenvalues whose eigenvectors are not orthogonal. A rotation by 90° is invertible and has no real eigenvalue at all.

2. **C** is correct. The trace of a correlation matrix is $N = 28$, so the share is $14.721/28 = 52.6\%$. A takes the eigenvalue λ_1 for a percentage. B divides by the sum of all 784 entries of C, $28 + 756 \times 0.50227 = 407.7$, the variance of the sum of the standardized returns, not their total variance.

3. **A** is correct. Covariance PCA swings toward the rescaled sector (Example 6). Correlations do not depend on units, so the eigenvalues (B) and loadings (C) of correlation PCA stay the same.

4. **A** is correct. The second half's $\mathbf{q}_2$ lies near the first half's plane (0.927) but not near the first half's $\mathbf{q}_2$ (0.297). So it is mostly the first half's $\mathbf{q}_3$ (cosine 0.879 in Section 7). The first half's $\mathbf{q}_2$ lies far from the second half's plane (0.514); it is mostly the second half's $\mathbf{q}_4$ (0.776). B reads the one-sided measure as two-sided: a plane that held would give both measures near 1. C is wrong: $|\cos\theta|$ ignores signs.

5. The trace is 3.6778, so $\frac{1}{2}\,\mathrm{tr} = 1.8389$. The term under the root is $\frac{1}{4}(1.3678)^2 + 0.7180^2 = 0.46772 + 0.51552 = 0.98324$, with root 0.99158, so $\lambda_1 = 2.8305$ and $\lambda_2 = 0.8473$. Then $x_2/x_1 = (2.8305 - 2.5228)/0.7180 = 0.4286$, an angle of 23.2°, so $\mathbf{q}_1 = (0.919,\ 0.394)^\top$. The share is $2.8305/3.6778 = 77.0\%$ and $\kappa = 2.8305/0.8473 = 3.34$.

6. (a) By Example 4 the eigenvalues are $1 + 4(0.5) = 3$ once and 0.5 four times; the first share is $3/5 = 60\%$ and $\kappa = 3/0.5 = 6$. (b) S5's first share is $2.8633/5 = 57.3\%$ and $\kappa = 2.8633/0.3362 = 8.52$: its smallest eigenvalue is below 0.5. At $\mathbf{u} = \mathbf{1}/\sqrt{5}$, $h = 1 + 4 \times 0.462 = 2.848 \le 2.8633$, as (11.4) requires.

7. With $\mathbf{q}_1 = (\cos 30°, \sin 30°)^\top$ and $\mathbf{q}_2 = (-\sin 30°, \cos 30°)^\top$, (11.3) gives $m_{11} = 4 \cdot \frac{3}{4} + \frac{1}{4} = 3.25$, $m_{22} = 4 \cdot \frac{1}{4} + \frac{3}{4} = 1.75$ and $m_{12} = 3 \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = 1.2990$. The trace is $5 = 4 + 1$, and with $m_{12}^2 = 27/16 = 1.6875$ the determinant is $3.25 \times 1.75 - 1.6875 = 4.0000 = 4 \times 1$.

8. Here $m_{11} - m_{22} = 1$, so $\tan 2\theta = 2\varepsilon$. For $\varepsilon = 0.1$, $\theta = \frac{1}{2}\arctan 0.2 = 5.65°$; to first order, 0.1 radians or $5.73°$. For $\varepsilon = 0.5$, $\theta = \frac{1}{2}\arctan 1 = 22.50°$; to first order, 0.5 radians or $28.65°$, 27% too large.

9. (a) Solving $(1 + \rho)/(1 - \rho) = 100$ gives $\rho = 99/101 = 0.9802$. (b) The condition number is $1.974365/0.025635 = 77.02$.

10. The bound is $10.863 \times 0.44235 = 4.805$, and the realized 0.910 is 0.19 of it; U28's realized 0.729 is only 0.024 of its 30.93. U28's κ is 11.6 times S5's, yet its solution moved less: the bound is a worst case, not a ranking.

11. The eigenvalues are 2.8633, 0.6674, 0.5895, 0.5436 and 0.3362, which add up to the trace, 5.0000. The first loadings are 0.441, 0.510, 0.444, 0.429 and 0.406, in S5's order. The scores have variances λ_k and covariances zero up to machine precision.

12. The year is 2022, with $\lambda_2 - \lambda_3 = 0.029$. Every year has $|\cos\theta_1| \geq 0.98$, and 2026 has the lowest, 0.981. The second cosine runs from 0.119 (2016) to 0.946 (2025), and the plane measure from 0.690 (2020) to 0.959 (2024). A high plane measure says only that the year's $\mathbf{q}_2$ lies near the full sample's plane, not that the year's plane matches it.

GLOSSARY

Term	中文	Meaning
Eigenvalue, eigenvector	特徵值、特徵向量	$M\mathbf{x} = \lambda_k \mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$: a direction that M only stretches, and the factor.
Characteristic equation	特徵方程式	$\det(M - \lambda_k I) = 0$, (11.1); for 2×2, (11.2).
Trace	跡	Sum of the diagonal entries; the sum of the eigenvalues.
Spectral theorem	譜定理	A real symmetric matrix has real eigenvalues and orthonormal eigenvectors.
Spectral decomposition	譜分解	$M = Q\Lambda Q^\top = \sum_k \lambda_k \mathbf{q}_k \mathbf{q}_k^\top$, (11.3).
Eigenspace	特徵空間	All eigenvectors of one eigenvalue, with $\mathbf{0}$.
Sign rule	符號規則	The book's convention: an eigenvector's entry largest in absolute value is positive.
Rayleigh quotient	Rayleigh 商	$h(\mathbf{u}) = \mathbf{u}^\top M\mathbf{u} / \mathbf{u}^\top\mathbf{u}$; between λ_N and λ_1, (11.4).
Condition number	條件數	$\kappa = \lambda_1 / \lambda_N$ for a positive definite matrix.
Spectral norm	譜範數	$\|M\|_2$, the largest length of $M\mathbf{u}$ over unit vectors.
Principal component analysis (PCA)	主成分分析	Directions of largest variance, one after another, (11.5).
Explained share	解釋比例	λ_k divided by the trace, (11.6).
Loading	負荷量	An entry of a principal component $\mathbf{q}_k$.
Score	主成分分數	$\xi_{k,t}$, a day's standardized returns read along $\mathbf{q}_k$.
Plane measure	平面度量	Cosine of the angle between one period's $\mathbf{q}_2$ and the plane of another period's $\mathbf{q}_2$ and $\mathbf{q}_3$; one-sided, so it does not show that the plane holds.
Stability check	穩定性檢查	The same calculation on two periods, compared; not an out-of-sample test.

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